

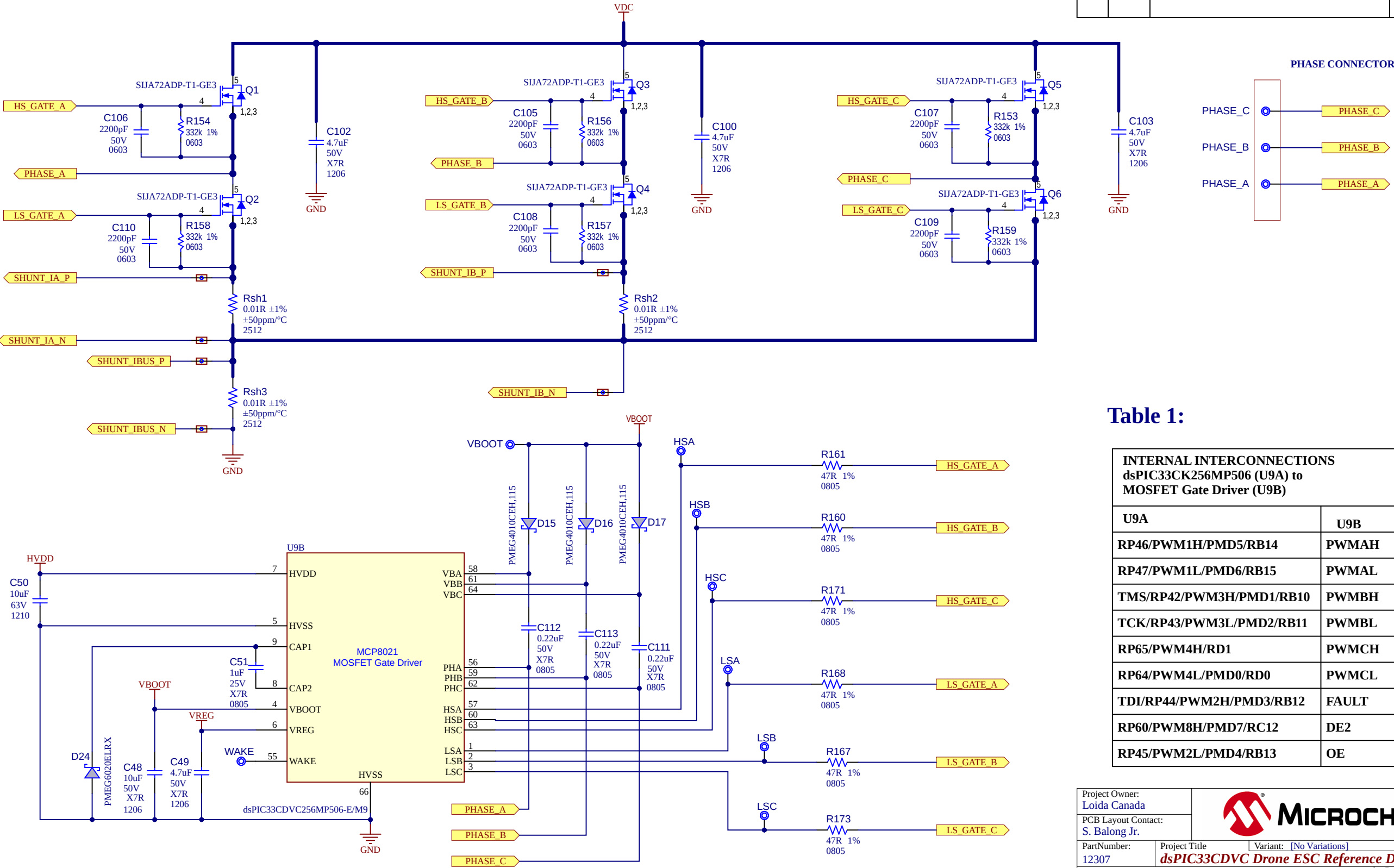
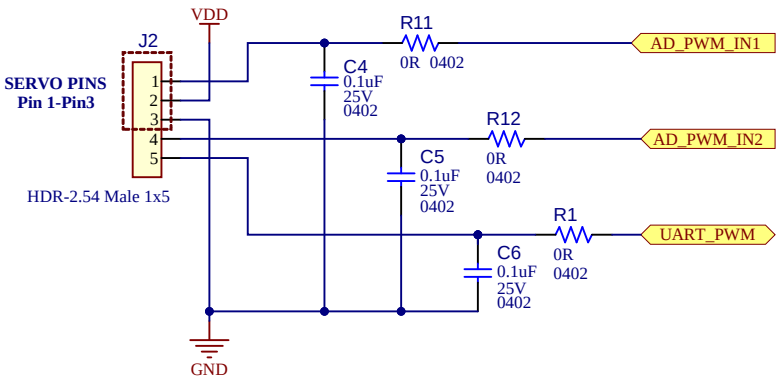


Table 1:

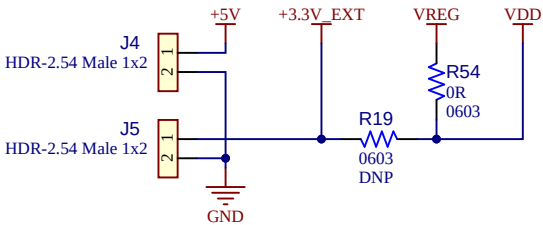
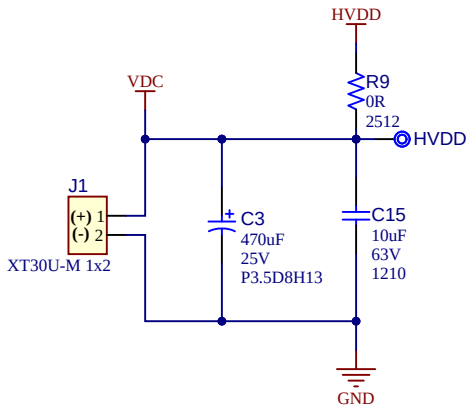
INTERNAL INTERCONNECTIONS dsPIC33CK256MP506 (U9A) to MOSFET Gate Driver (U9B)		
U9A	U9B	
RP46/PWM1H/PMD5/RB14	PWMAH	
RP47/PWM1L/PMD6/RB15	PWMAL	
TMS/RP42/PWM3H/PMD1/RB10	PWMBH	
TCK/RP43/PWM3L/PMD2/RB11	PWMBL	
RP65/PWM4H/RD1	PWMCH	
RP64/PWM4L/PMD0/RD0	PWMCL	
TDI/RP44/PWM2H/PMD3/RB12	FAULT	
RP60/PWM8H/PMD7/RB13	DE2	
RP45/PWM2L/PMD4/RB13	OE	

REV	ECO#	DESCRIPTION	DATE

External PWM/ADC Control



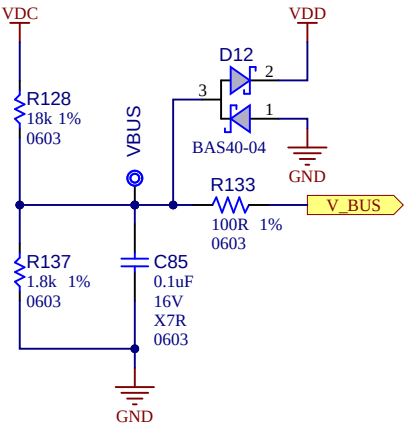
Power Supply



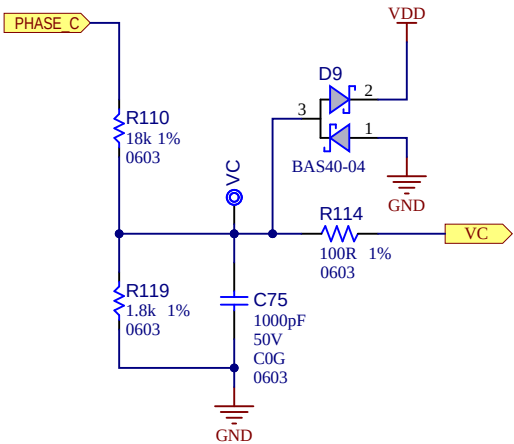
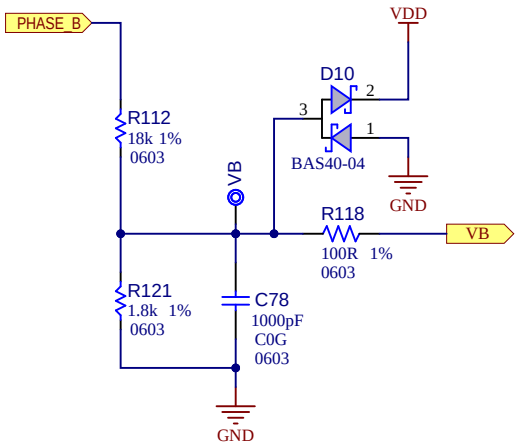
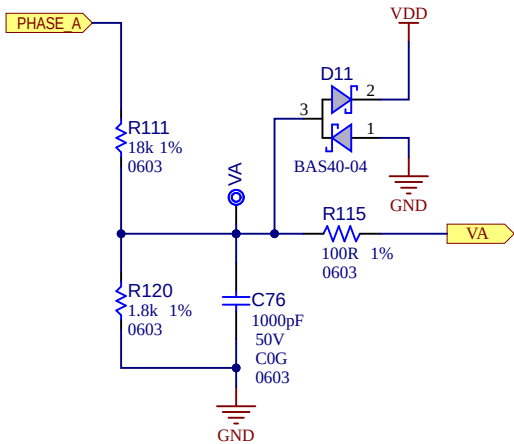
Ensure total power dissipation of the dsPIC33CDVC256MP506 (U9) does not exceed the specified value when internal 3.3V LDO (VREG- U9/U9B PIN#6) is used for powering dsPIC33CK256MP506 (U9A) by populating R54 instead of R19.

Resistor Configuration for the Supply Voltages:
+5V : from External Source, to power the CAN Transceiver
VREG : R19=DNP and R54=0R, +3.3V is derived from Internal VREG (U9/U9B Pin #6)
: In this configuration, do not draw more than 20mA
+3.3V_EXT : When R19=0R, R54=DNP, 3.3V is derived from External Source

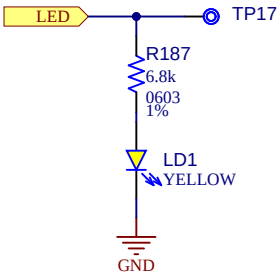
DC Bus Voltage Sensing



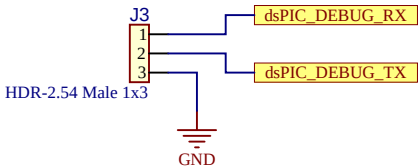
Phase Voltage Sensing



General Purpose LEDs



UART Communication Header



Project Owner: Loida Canada			
PCB Layout Contact: S. Balong Jr.			
PartNumber: 12307	Project Title dsPIC33CDVC Drone ESC Reference Design	Variant: [No Variations]	
Sheet Title Program/Debug Interface + Sensing/Feedback Ckts		Designed with 	
Size B	SCH #: 03-01243	Rev: 2	Date: 1/20/2026
	PCB #: 04-12307	Rev: 2	Sheet 3 of 4
File: Sheet3_SenseCkts_Programming_Interface.SchDoc			

